Proposition 5: the lower bound $4/3$ for the edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition5_EditLowerBound` formalises Proposition 5 (p. 6) with the construction of Appendix C.1, Lemmas 23–25 and Claim 26, and Theorem 3 (1). Shared notions and lemmas (the edit distance, Observation 21, Lemma 22) come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Prop5.proposition5_dEdit`: for every odd $N \geq 3$, the graphs $G =$ `GN N` and $H =$ `HN N` on the common vertex set `VN N` satisfy

$$d_{\mathrm{ed}}(G,H) \geq 4N-2, \qquad d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq 3N+1;$$

`GsimOT.Prop5.proposition5` states it for the costs ($C_{\mathrm{ed}}(\pi) \geq 4N-2$ for all bijections π, $C_{\mathrm{ed}}(Q) \leq 3N+1$ for some $Q \in \mathrm{DS}(V,W)$). `GsimOT.Prop5.proposition5_delta` (and `GsimOT.Prop5.theorem3_item1`, Theorem 3 (1)): for every $\varepsilon > 0$ there is an odd $N \geq 3$ with $\delta_{\mathrm{ed}}(G,H) \geq (\frac{4}{3}-\varepsilon)\,\delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$; by `GsimOT.Prop5.proposition5_delta_of_le` this holds for every odd $N \geq 3$ with $\frac{10}{3(3N+1)} \leq \varepsilon$.

Correspondence: Proposition 5 and its proof

Paper	Page	Statement	Lean declaration(s) and location
Appendix C.1 (construction, l. 677–687)	21	Markers $z_0, \dots, z_{N-1}$; cores $S_G = \{v_0, v_2, \dots, v_{2N-2}\}$, $S_H = \{w_1, \dots, w_{2N-1}\}$ at their positions in $\mathbb{Z}_{2N}$; distinct core vertices adjacent iff their positions sum to 0 mod $2N$; z_r adjacent to the core vertices on $I_r = \{(r+s) \bmod 2N \mid s < N\}$	`GsimOT.Prop5.Base` Proposition5_EditLowerBound/Construction.lean:37 `GsimOT.Prop5.semicircle` Proposition5_EditLowerBound/Construction.lean:40 `GsimOT.Prop5.mem_semicircle` Proposition5_EditLowerBound/Construction.lean:43 `GsimOT.Prop5.posG` Proposition5_EditLowerBound/Construction.lean:53 `GsimOT.Prop5.posH` Proposition5_EditLowerBound/Construction.lean:56 `GsimOT.Prop5.coreGraph` Proposition5_EditLowerBound/Construction.lean:68 `GsimOT.Prop5.Gp` Proposition5_EditLowerBound/Construction.lean:88 `GsimOT.Prop5.Hp` Proposition5_EditLowerBound/Construction.lean:92
Appendix C.1 (construction, l. 692–694)	22	For odd N: core edges $\{c, c'\}$ with $c + c' = N$ in G' and with $c + c' + 1 = N$, $c \neq c'$ in H'; v_0 and w_N are isolated	`GsimOT.Prop5.Gp_adj_inl_inl` Proposition5_EditLowerBound/Construction.lean:106 `GsimOT.Prop5.Hp_adj_inl_inl` Proposition5_EditLowerBound/Construction.lean:117

Paper	Page	Statement	Lean declaration(s) and location

Appendix C.1 (construction, l. 688–691)	21–22	G, H by Lemma 22, markers ordered by their degree in G', $\Delta = N + 1$, $\tau = 4N - 1$, $\delta = 1$; all degrees are at most $N + 1$, and the degrees of z_r in G' and H' differ by one	`GsimOT.Prop5.markerEnum`
			`Proposition5_EditLowerBound/Construction.lean:249`
			`GsimOT.Prop5.markerEnum_mono`
			`Proposition5_EditLowerBound/Construction.lean:254`
			`GsimOT.Prop5.deg_Gp_le`
			`Proposition5_EditLowerBound/Construction.lean:218`
			`GsimOT.Prop5.deg_Hp_le`
			`Proposition5_EditLowerBound/Construction.lean:231`
			`GsimOT.Prop5.deg_Gp_inr`
			`Proposition5_EditLowerBound/Construction.lean:189`
			`GsimOT.Prop5.deg_Hp_inr`
			`Proposition5_EditLowerBound/Construction.lean:200`
			`GsimOT.Prop5.markerEnum_deg`
			`Proposition5_EditLowerBound/Construction.lean:262`
			`GsimOT.Prop5.mK`
			`Proposition5_EditLowerBound/Construction.lean:270`
			`GsimOT.Prop5.GN`
			`Proposition5_EditLowerBound/Construction.lean:276`
			`GsimOT.Prop5.HN`
			`Proposition5_EditLowerBound/Construction.lean:279`

Lemma 23, Fig. 4 (l. 696–699, 703–706)	22	$\ell(v_i, w_j) = \min_k \lvert i - j + 2kN \rvert$; mapping v_i to w_j changes the adjacency to exactly $\ell(v_i, w_j)$ markers	`GsimOT.Prop5.cycDist`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:37`
			`GsimOT.Prop5.ell`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:73`
			`GsimOT.Prop5.card_semicircle_ne`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:51`
			`GsimOT.Prop5.sum_marker_inl`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:77`
			`GsimOT.Prop5.sum_marker_inr`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:82`

Paper	Page	Statement	Lean declaration(s) and location

Lemma 23 (l. 700–702)	22	For compliant Q, the core-marker pairs contribute exactly $2\sum Q(v_i, w_j)\,\ell(v_i, w_j)$ to $C_{\mathrm{ed}}(Q)$	`GsimOT.Prop5.extMarkers`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:90`
			`GsimOT.Prop5.coreCost`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:120`
			`GsimOT.Prop5.costEdit_extMarkers`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:127`
			`GsimOT.Prop5.lemma23`
			`Proposition5_EditLowerBound/Lemma23_MarkerCost.lean:165`

Proof of Lemma 24 (l. 708–714)	22	$\pi_{\pm}(v_i) = w_{i\pm1}$, $Q = \frac{1}{2}(P_{\pi_-} + P_{\pi_+})$; $\ell(v_i, \pi_{\pm}(v_i)) = 1$	`GsimOT.Prop5.predCore`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:37`
			`GsimOT.Prop5.avgPerm`
			`Proposition5_EditLowerBound/CoreEdges.lean:170`
			`GsimOT.Prop5.Rwit`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:106`
			`GsimOT.Prop5.Qwit`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:109`
			`GsimOT.Prop5.ell_self`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:116`
			`GsimOT.Prop5.ell_predCore`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:122`

Proof of Lemma 24, Fig. 5 (l. 715–724)	22–23	Core-core pairs: the core edges are preserved iff one endpoint moves forward and the other backward, except near v_0 and w_N; the Lean computes their cost as $\frac{1}{4}(2(N-1)+2+2+2(N-1)) = N$	`GsimOT.Prop5.pres_eq`
			`Proposition5_EditLowerBound/CoreEdges.lean:135`
			`GsimOT.Prop5.pres_id_id`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:57`
			`GsimOT.Prop5.pres_pred_pred`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:65`
			`GsimOT.Prop5.pres_id_pred`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:75`
			`GsimOT.Prop5.pres_pred_id`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:90`
			`GsimOT.Prop5.coreCost_avgPerm`
			`Proposition5_EditLowerBound/CoreEdges.lean:210`
			`GsimOT.Prop5.coreCost_witness`
			`Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:128`

Paper	Page	Statement	Lean declaration(s) and location
Lemma 24 (l. 707, 725)	22–23	$d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \le 3N+1$, from $C_{\mathrm{ed}}(Q) = 3N$ for the witness Q	`GsimOT.Prop5.costEdit_witness` Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:139 `GsimOT.Prop5.lemma24` Proposition5_EditLowerBound/Lemma24_FractionalCost.lean:145
Proof of Lemma 25 (l. 727–729)	23	If $C_{\mathrm{ed}}(\pi) < \tau$, then π fixes the markers and maps S_G onto S_H (Lemma 22); the pairs involving the clique only add to the cost	`GsimOT.lemma22_bij_edit` Common/Lemma22_LockingMarkersEdit.lean:41 `GsimOT.Prop5.restrictInl` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:44 `GsimOT.Prop5.costEditBij_restrict` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:56 `GsimOT.Prop5.costEditBij_sumCongr` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:76
Proof of Lemma 25, Claim 26 (l. 729–742)	23–24	Displacements $d_i \in \{-N+2, \dots, N\}$ with $\ell(v_i, \pi(v_i)) = \lvert d_i \rvert$; $\sum d_i \equiv N \pmod{2N}$, so $\lvert \sum d_i \rvert \ge N$	`GsimOT.Prop5.disp` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:86 `GsimOT.Prop5.ell_eq_abs_disp` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:92 `GsimOT.Prop5.one_le_ell` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:99 `GsimOT.Prop5.sum_disp` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:115 `GsimOT.Prop5.sum_disp_ge` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:125
Claim 26 (l. 737–749)	24	$U \ge N + 2k$; preserved edges have $d_i + d_{i'} = 0$	`GsimOT.Prop5.Preserved` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:136 `GsimOT.Prop5.disp_add_disp_partner` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:154 `GsimOT.Prop5.claim26` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:167

Paper	Page	Statement	Lean declaration(s) and location
Proof of Lemma 25 (l. 750–751)	24	Both cores have $(N-1)/2$ edges; their ordered pairs contribute $2(N-1-2k)$	`GsimOT.Prop5.sum_adj_Gp_core` Proposition5_EditLowerBound/CoreEdges.lean:96 `GsimOT.Prop5.sum_adj_Hp_core` Proposition5_EditLowerBound/CoreEdges.lean:108 `GsimOT.Prop5.mixCore` Proposition5_EditLowerBound/CoreEdges.lean:125 `GsimOT.Prop5.pres` Proposition5_EditLowerBound/CoreEdges.lean:130 `GsimOT.Prop5.mixCore_eq` Proposition5_EditLowerBound/CoreEdges.lean:145
Lemma 25 (l. 726, 753)	23–24	$C_{\mathrm{ed}}(\pi) \geq 4N-2$ for every π, i.e. $d_{\mathrm{ed}}(G,H) \geq 4N-2$	`GsimOT.Prop5.lemma25_bij` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:237 `GsimOT.Prop5.lemma25` Proposition5_EditLowerBound/Lemma25_BijectionCost.lean:275
Proposition 5 (l. 221–222)	6	$d_{\mathrm{ed}}(G,H) \geq 4N-2$ and $d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq 3N+1$; in terms of costs: $C_{\mathrm{ed}}(\pi) \geq 4N-2$ for all π, $C_{\mathrm{ed}}(Q) \leq 3N+1$ for some Q	`GsimOT.Prop5.proposition5_dEdit` Proposition5_EditLowerBound/Proposition5.lean:31 `GsimOT.Prop5.proposition5` Proposition5_EditLowerBound/Proposition5.lean:38
Proof of Proposition 5 (l. 754–758)	24	$d_{\mathrm{ed}} \geq 4N-2 = (\frac{4}{3} - \frac{10}{3(3N+1)})(3N+1) \geq (\frac{4}{3} - \frac{10}{3(3N+1)})\, d_{\mathrm{ed}}^{\mathrm{OT}}$; same normalisation; choose N with $\frac{10}{3(3N+1)} \leq \varepsilon$	`GsimOT.Prop5.proposition5_delta_of_le` Proposition5_EditLowerBound/Proposition5.lean:45
Proposition 5 (l. 222–223)	6	For every $\varepsilon > 0$ there is an N with $\delta_{\mathrm{ed}}(G,H) \geq (\frac{4}{3} - \varepsilon)\, \delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$	`GsimOT.Prop5.proposition5_delta` Proposition5_EditLowerBound/Proposition5.lean:68
Theorem 3 (1) (l. 124)	4	For every $\varepsilon > 0$ some G, H have $\delta_{\mathrm{ed}} \geq (\frac{4}{3} - \varepsilon)\, \delta_{\mathrm{ed}}^{\mathrm{OT}}$	`GsimOT.Prop5.theorem3_item1` Proposition5_EditLowerBound/Proposition5.lean:82

Shared notions and lemmas used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Section 2	2, 5	$C_{\mathrm{ed}}(Q) = \sum_{v,w,v',w'} \lvert D(v,w,v',w')\rvert\, Q(v,w)\, Q(v',w')$ and $C_{\mathrm{ed}}(\pi) = C_{\mathrm{ed}}(P_\pi) = \lVert A_G^\pi - A_H\rVert_{\mathrm{ed}}$	`GsimOT.costEdit` Common/EditDistance.lean:35 `GsimOT.costEditBij` Common/EditDistance.lean:39 `GsimOT.costEditBij_eq` Common/EditDistance.lean:60
Section 1, Eq. (1), Section 2, Appendix A	2, 5, 14–15	$d_{\mathrm{ed}} = \min_\pi C_{\mathrm{ed}}(\pi)$ and $d_{\mathrm{ed}}^{\mathrm{OT}} = \min_{Q\in \mathrm{DS}(V,W)} C_{\mathrm{ed}}(Q)$ (infimum, attained); $d_{\mathrm{ed}}^{\mathrm{OT}} \le d_{\mathrm{ed}}$	`GsimOT.dEdit` Common/EditDistance.lean:42 `GsimOT.dOTEdit` Common/EditDistance.lean:45 `GsimOT.exists_dOTEdit_eq` Common/EditDistance.lean:109 `GsimOT.dOTEdit_le_dEdit` Common/EditDistance.lean:102
Section 1, Section 2	1–2, 5	$\delta_{\mathrm{ed}} = n^{-2} d_{\mathrm{ed}}$, $\delta_{\mathrm{ed}}^{\mathrm{OT}} = n^{-2} d_{\mathrm{ed}}^{\mathrm{OT}}$ (scaling by $1/n^2$)	`GsimOT.deltaEdit` Common/EditDistance.lean:48 `GsimOT.deltaOTEdit` Common/EditDistance.lean:51
Section 1, Section 2, Appendix A	2, 5, 14	$C_{\mathrm{ed}}(Q) = \lVert B(G,H,Q)\rVert_{\mathrm{ed}}$ for $Q \ge 0$ (entrywise 1-norm $\lVert\cdot\rVert_{\mathrm{ed}} = \lVert\cdot\rVert_{(1)}$)	`GsimOT.entrywiseNorm` Common/MatrixNorms.lean:53 `GsimOT.costEdit_eq_entrywiseNorm_B` Common/MatrixNorms.lean:119
Section 2, Eq. (3)	5	Birkhoff–von Neumann: $Q = \sum_i \lambda_i P_{\pi_i}$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (also: over all $\sigma \in \mathrm{Perm}(V)$ with $\lambda \ge 0$); conversely every such combination is in $\mathrm{DS}(V,W)$	`GsimOT.exists_birkhoff_decomposition_pos` Common/BirkhoffDecomposition.lean:72 `GsimOT.exists_birkhoff_decomposition` Common/BirkhoffDecomposition.lean:42 `GsimOT.isDS_of_decomposition` Common/BirkhoffDecomposition.lean:23
Observation 21	20	$L_Q(v,w) \ge \lvert \deg_G(v) - \deg_H(w)\rvert$	`GsimOT.observation21` Common/Observation21_DegreeBound.lean:21 `GsimOT.observation21_bij` Common/Observation21_DegreeBound.lean:40

Paper	Page	Statement	Lean declaration(s) and location
Lemma 22 (construction)	21	Adding the clique K ($m = 2\Delta + 2\tau + k(\tau+\delta)+1$) and the staircase	`GsimOT.lockSize` Common/Lemma22_LockingMarkers.lean:36 `GsimOT.stair` Common/Lemma22_LockingMarkers.lean:40 `GsimOT.lockGraph` Common/Lemma22_LockingMarkers.lean:56
Lemma 22 (2)	21	$Q(z_j, z_j) = 1 \Rightarrow C_1(Q')$ w.r.t. (G', H') equals $C_1(Q)$ w.r.t. (G, H)	`GsimOT.extendId` Common/Lemma22_LockingMarkers.lean:260 `GsimOT.lemma22_cost` Common/Lemma22_LockingMarkers.lean:351
Lemma 22 (1), $\diamondsuit$ = ed	21	$C_{\mathrm{ed}}(\pi) < \tau \Rightarrow \pi$ fixes all markers and $\pi(V) = V$; via $C_1(\pi) \le C_{\mathrm{ed}}(\pi)$ and the case $\diamondsuit = 1$	`GsimOT.costOneBij_le_costEditBij` Common/Lemma22_LockingMarkersEdit.lean:26 `GsimOT.lemma22_bij_edit` Common/Lemma22_LockingMarkersEdit.lean:41
Lemma 22 (2), $\diamondsuit$ = ed	21	$Q(z_j, z_j) = 1 \Rightarrow C_{\mathrm{ed}}(Q')$ w.r.t. (G', H') equals $C_{\mathrm{ed}}(Q)$ w.r.t. (G, H)	`GsimOT.lemma22_cost_edit` Common/Lemma22_LockingMarkersEdit.lean:53

Notes on the formalisation

1. Proposition 5 states $d_{\mathrm{ed}}(G,H) \ge 4N-2$ and $d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \le 3N+1$ (`proposition5_dEdit`); `proposition5` is the form with the explicit witness Q. `proposition5_delta_of_le` is the computation of the proof for every odd $N \ge 3$ with $\frac{10}{3(3N+1)} \le \varepsilon$.

2. The common vertex set of G' and H' is `Base N` $=$ `Fin N` $\oplus$ `Fin N`: `Sum.inl c` is the core vertex v_{2c} of G' and w_{2c+1} of H' (the identification $v_i \leftrightarrow w_{i+1}$ of the paper), `Sum.inr r` is the marker z_r, and $\mathbb{Z}_{2N}$ is $\{0, \dots, 2N-1\}$. With this identification, π_+ is the identity on the core and π_- maps the core vertex c to $c-1 \bmod N$ (`predCore`).

3. A compliant Q is `extendId (extMarkers R)`, with R the restriction of Q to the core vertices. Lemma 23 is computed on (G', H') (`costEdit_extMarkers`) and transferred to (G, H) by Lemma 22 (2) (`lemma23`).

4. Lemma 24: `costEdit_witness` computes the cost of the witness, $C_{\mathrm{ed}}(Q) = 3N$, and `lemma24` derives $d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \le 3N+1$ from it.

5. Lemma 25: instead of replacing π on the clique by the identity, the pairs involving the clique are dropped from $C_{\mathrm{ed}}(\pi)$ (`costEditBij_restrict`). Claim 26 is stated with $2k$, the number of preserved ordered core edges (`pres`), and $\sum_{v_i \in S} d_i = 0$ is proved with the involution $c \mapsto N - c$ on S.

6. Lemma 22 for $\diamondsuit$ = ed is in `Common/Lemma22_LockingMarkersEdit.lean`; its first item follows from the case $\diamondsuit = 1$, as $C_1(\pi) \le C_{\mathrm{ed}}(\pi)$. The parameters $k = N$, $\Delta = N+1$, $\tau = 4N-1$, $\delta = 1$ are those of the paper (`mK`, `GN`).